

$$1. \text{ VAE } E_z \sim q_\phi(z|x) [\log P_\theta(x|z)] - D_{KL}(q_\phi(z|x), p(z))$$

$$\text{GAN } \min_G \max_D [E_x [\log D(x)] + E_z [\log (1 - D(G(z)))]]$$

VAE optimizes variational lower bound
the reconstruction loss helps the model to output info like the original image, but it averages all the plausible outputs, so this leads to blurriness.

GAN trains generator to compete with discriminator. The generator helps produce image that look more real, so the images are clearer.

2. The discriminator judges if the output is good enough. It helps distinguish between images that look real and images that look fake. The generator's goal is to produce output that passes the requirement of the discriminator.